上海交通大学学位论文

面向 3C 工业生产线的双臂机器人视觉-语言-动作模型

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A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES

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摘 要

现代 3C 电子装配线需要处理小型元器件、高产品多样性和富接触操作，这些任务仍难以通过传统固定程序机器人实现自动化。本项目针对这一缺口，面向双臂工业操作开发视觉-语言-动作（Vision-Language-Action, VLA）系统。系统设计运行于双臂 Franka 平台，并面向典型 PC 关键部件任务，包括内存条插入、CPU 放置以及电源线连接。

当前部署系统选择相应的 ACT 检查点，接收同步视觉观测和机器人状态，并在闭环执行过程中输出机器人及夹爪动作。现阶段证据限于干运行和有边界的验证活动；本文不声称已经实现语言条件推理、统一多任务控制、自主规划、故障检测、恢复或双臂协同操作。这些功能仍是面向预期 VLA 与双臂系统的长期扩展方向。

工程验收目标包括：规划成功率高于 90%，任务执行成功率高于 70%，部件检测精度高于 90%，位姿误差低于 20 mm，故障检测率高于 90%，恢复成功率高于 70%。这些数值定义验证阈值，而非已经证明的性能结果。

关键词： 双臂机器人，视觉-语言-动作模型，工业自动化

ABSTRACT

Modern 3C electronics assembly lines deal with small components, high product variety, and contact-rich operations that remain difficult to automate with conventional fixed-program robots. Our project targets this gap by developing a Vision-Language-Action (VLA) system for bimanual industrial manipulation. The system is designed to run on a dual-arm Franka platform and to handle representative PC key-component tasks—specifically RAM insertion, CPU placement, and power-cable connection.

The deployed system selects the corresponding ACT checkpoint, receives synchronized visual observations and robot state, and outputs robot and gripper actions during closed-loop execution. Current evidence is limited to dry runs and bounded validation activities; the report does not claim demonstrated language-conditioned inference, unified multi-task control, autonomous planning, failure detection, recovery, or bimanual manipulation. These functions remain long-term extensions toward the intended Vision-Language-Action (VLA) and dual-arm system.

The engineering acceptance targets are planning success above 90%, task execution success above 70%, component detection accuracy above 90%, pose error below 20 mm, failure-detection rate above 90%, and recovery success above 70%. These values define validation thresholds rather than demonstrated performance results.

Key words: dual-arm robot, vision-language-action model, industrial automation

Contents

Chapter 1 Introduction

1.1 Background and significance

The 3C industry—computer, communication, and consumer electronics—increasingly demands flexible robotic systems that can cope with short product life cycles, diverse part geometries, small components, and frequent line changeovers. Traditional hard-coded automation approaches are effective for stable, high-volume production, but they exhibit significant limitations in high-mix assembly scenarios. Each product variant typically requires reprogramming, fixture redesign, and system reconfiguration, which increases development and maintenance costs. In contrast, human operators can rapidly understand new tasks from natural-language instructions and adapt to new operating conditions with limited prior programming. This contrast highlights the importance of advancing industrial robotic systems toward more human-aligned capabilities, especially semantic understanding of instructions and data-efficient adaptation under limited demonstrations.

Recent advances in Vision-Language-Action (VLA) models offer a promising route toward this goal. By jointly learning from visual observations, language instructions, robot proprioceptive states, and action data, VLA models can in principle generalize across tasks without requiring separate hand-engineered programs for every product variant. Landmark systems, including RT-2, OpenVLA, RDT-1B, and the π series, demonstrate the potential of transferring semantic knowledge from large vision-language models to physical robot control[rt2,openvla,rdt1b,pi0]. Earlier work on visual and language-conditioned manipulation also established complementary design patterns: Transporter Networks use spatially grounded pick-and-place representations, CLIPort combines semantic and spatial pathways, and PerAct couples language goals to structured visual observations[transporter,cliport,peract]. In parallel, BridgeData V2, Open X-Embodiment, DROID, and LIBERO have made dataset scale, embodiment diversity, and reproducible task evaluation central considerations in robot-learning system design[bridgev2,openx,droid,libero].

Despite this progress, adapting VLA-style systems to industrial assembly remains a system-level engineering problem. Many published demonstrations emphasize tabletop manipulation, single-arm control, or short-horizon tasks with relatively forgiving tolerances. In contrast, 3C assembly requires reliable perception of small components, accurate contact-rich manipulation, synchronized sensing, safe robot execution, and repeatable validation under bounded physical conditions. These requirements motivate a complete manipulation pipeline that links hardware selection, teleoperation, data collection, model training, deployment, and quantitative evaluation.

1.2 Design problem

This project aims to prototype a dual-arm industrial manipulation platform that couples camera observations and robot proprioceptive state to learned task actions for a scoped set of PC assembly operations. The target workcell consists of two Franka Research 3 (FR3) robot arms with complementary global and wrist/side camera views. Demonstrations are collected through a GELLO-based teleoperation interface and are recorded as synchronized visual observations, robot states, and robot or gripper actions. The intended task domain covers representative PC key-component operations, including RAM insertion, CPU placement, and power-cable connection.

At the present stage, task-specific Action Chunking with Transformers (ACT) checkpoints are selected by the operator rather than inferred from a language command[act]. The deployed system receives synchronized observations and robot state, then outputs robot and gripper actions for closed-loop execution. This implementation is therefore best understood as a staged imitation-learning baseline and a foundation for the intended VLA system, rather than as a fully demonstrated language-conditioned, unified multi-task controller. Language-conditioned inference, autonomous planning, failure detection, recovery, and robust bimanual coordination remain long-term extensions toward the final VLA and dual-arm manipulation concept.

1.3 Project objectives and scope

The project objective is to establish an end-to-end engineering pipeline for bounded industrial manipulation experiments. This pipeline includes hardware integration, synchronized multi-camera sensing, teleoperated demonstration collection, dataset preparation, ACT-based policy training, policy deployment, and repeatable physical validation. The expected

Design Expo outcome is a measurable real-robot demonstration of the scoped assembly step, including task success, pose-estimation performance, and failure-handling evaluation.

The engineering acceptance targets are planning success above 90%, task execution success above 70%, component detection accuracy above 90%, pose error below 20 mm, failure-detection rate above 90%, and recovery success above 70%. These values define validation thresholds rather than demonstrated performance results. Current evidence is limited to dry runs and bounded validation activities; therefore, this thesis does not claim unrestricted open-world generalization, autonomous factory deployment, guaranteed safe recovery from every failure mode, or fully demonstrated VLA-based multi-task inference.

1.4 Main contributions

The main contributions of this thesis are as follows:

1. A dual-FR3 manipulation platform is specified and integrated with GELLO teleoperation, synchronized camera sensing, robot-state recording, and task-compatible end effectors and fixtures.
2. A reproducible data and learning pipeline is developed to convert teleoperated demonstrations into datasets for ACT-based training and real-robot deployment.
3. A scoped industrial assembly validation framework is defined for representative PC component tasks, with quantitative targets for planning, execution, perception, pose accuracy, failure detection, and recovery.
4. A traceable engineering process links customer requirements and quality function deployment to concept generation, module selection, implementation decisions, and validation criteria.

1.5 Organization of this thesis

The remainder of this thesis is organized as follows. Chapter 4 translates customer requirements into measurable engineering specifications and presents the quality function deployment. Chapter 6 describes functional decomposition, concept generation, and selection of the major hardware, sensing, teleoperation, end-effector, and policy modules. Chapter 7 details the final system design, data and control architecture, model-training procedure, and implementation plan. Chapter 9 presents the physical validation methods and experimental results. Chapter 11 discusses the strengths and limitations of the system, summarizes the conclusions, and identifies directions for future work.

Chapter 2 Introduction

2.1 Background and significance

The 3C industry—computer, communication, and consumer electronics—increasingly demands flexible robotic systems that can cope with short product life cycles, diverse part geometries, small components, and frequent line changeovers. Traditional hard-coded automation approaches are effective for stable, high-volume production, but they exhibit significant limitations in high-mix assembly scenarios. Each product variant typically requires reprogramming, fixture redesign, and system reconfiguration, which increases development and maintenance costs. In contrast, human operators can rapidly understand new tasks from natural-language instructions and adapt to new operating conditions with limited prior programming. This contrast highlights the importance of advancing industrial robotic systems toward more human-aligned capabilities, especially semantic understanding of instructions and data-efficient adaptation under limited demonstrations.

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Chapter 3 Design specification

3.1 Customer requirements

Through discussions with the project sponsor and analysis of the target application, eight customer requirements were identified and assigned importance weights reflecting the sponsor's priorities. The requirements capture both functional goals and non-functional concerns, including robustness, safety, generalizability, cost, and reliability.

Table 3–1 Customer requirements and importance weights

No.	Customer Requirement	Weight
CR1	Complete assembly task successfully	0.18
CR2	High assembly precision	0.16
CR3	Robust to environment and disturbances	0.15
CR4	Safe and collision-free operation	0.14
CR5	Fast task execution	0.12
CR6	Easy to use and generalize	0.10
CR7	Cost-effective system	0.08
CR8	System reliable and recovery capable	0.07

The customer requirements were translated into engineering specifications through the QFD process rather than by assigning targets independently. For each customer requirement i and engineering specification j, the team assigned a relationship value $r_{ij} \in \{0, 1, 3, 9\}$ (none, weak, moderate, or strong) and calculated the unnormalized engineering priority as

$$I_j = \sum_i w_i r_{ij}, \tag{3–1}$$

where w_i is the customer-importance weight in Table 4–1. The priorities were then normalized to identify the technical characteristics that most directly protect the high-weight needs. CR1 and CR5 primarily drive ES1 and ES2; CR2 drives ES3 and ES4; CR3, CR4, and CR8 drive ES5 and ES6; and CR6–CR7 constrain the selected implementation to a usable, feasible, and maintainable route. This translation produces an auditable chain from the sponsor's language to acceptance tests and explains why the later concept search is organized around platform, teleoperation, data collection, gripper, and policy modules.

3.2 Engineering specifications

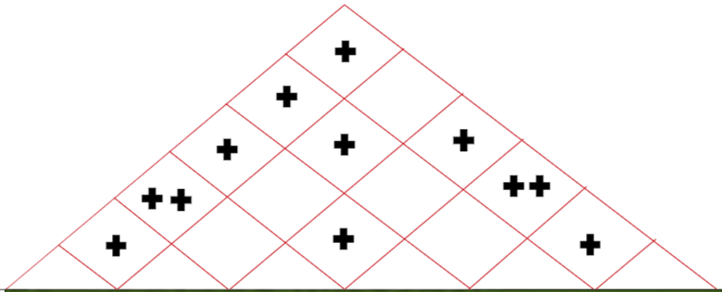
The customer requirements were converted into six measurable engineering specifications. The targets in Table 4–2 are the final acceptance thresholds for the present proof-of-concept, rather than performance claims. Each rate is calculated over independently initialized physical trials of the relevant task. A trial is counted once only; an unsuccessful recovery is not reclassified as an independent execution attempt. This convention prevents recovery behavior from artificially inflating the reported task-execution success rate.

Table 3–2 Final engineering specifications and acceptance thresholds

No.	Engineering Specification	Target	Operational Definition
ES1	Planning success rate	$> 90\%$	Fraction of trials in which the selected task policy and initialized scene form a valid executable plan.
ES2	Task execution success rate	$> 70\%$	Fraction of trials in which the prescribed assembly operation reaches its task-specific completion condition.
ES3	Component detection accuracy	$> 90\%$	Fraction of evaluated observations in which the required component is correctly detected for the active task.
ES4	Pose estimation error	$< 20\,\mathrm{mm}$	Euclidean position error between the estimated and reference component pose.
ES5	Failure detection rate	$> 90\%$	Fraction of induced or naturally occurring execution failures that are identified before further unsafe action is issued.
ES6	Recovery rate	$> 70\%$	Fraction of detected failures for which the defined recovery procedure restores a valid task state.

3.3 Quality function deployment

Figure 4–1 records the resulting House of Quality. Its 9–3–1 relationship scale makes the engineering priorities interpretable rather than decorative: task completion, component localization, pose estimation, and failure handling carry influence across multiple high-weight customer requirements. The roof is used as a design-control device. For example, additional visual coverage can strengthen ES3 and ES4, but it increases calibration, synchronization, and data-management burden; similarly, recovery logic can improve ES5–ES6 while adding controller integration work. These QFD trade-offs define the subsequent concept space. The platform and gripper must enable precise contact motion for ES1–ES2 and ES4; the tele-operator and data pipeline must produce usable demonstrations and observable scenes; and the policy must convert those observations into safe executable actions while leaving explicit hooks for failure monitoring and recovery.



Customer Requirements	Weight (Importance)	Engineering Specifications (ES)					
		1. Task Execution Success Rate (%)	2. Pose Error (mm)	3. Failure Detection Rate (%)	4. Recovery Success Rate (%)	5. Component Detection Accuracy (%)	6. Planning Success Rate (%)
Complete assembly task successfully	0.18	9	3	3	3	3	9
High assembly precision	0.16	3	9	1	1	9	3
Robust to environment and disturbances	0.15	3	3	9	9	3	1
Safe operation	0.14	3	1	3	3	1	1
Fast task execution	0.12	3	1	1	1	1	3
Easy to use and generalize	0.10	1	3	3	3	3	9
Cost-effective system	0.08	1	1	1	1	1	1
System reliable and recovery capable	0.07	3	1	9	9	1	3
Absolute Importance (Sum)		372	314	360	360	314	394
Relative Importance (%)		17.6	14.9	17.0	17.0	14.9	18.6
Rank (1 = Highest)		2	5	3	3	5	1
Benchmark		85%	3.2 mm	85%	75%	92%	88%
Target		>70%	<20 mm	>90%	>70%	>90%	>90%

Illustration 3–1 QFD House of Quality for the VLA bimanual manipulation system.

Chapter 4 Design specification

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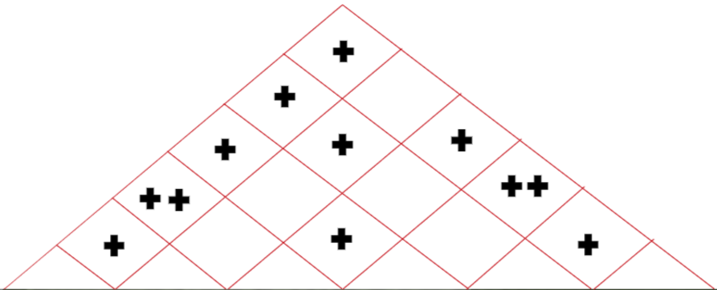

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Benchmark		85%	3.2 mm	85%	75%	92%	88%
Target		>70%	<20 mm	>90%	>70%	>90%	>90%

Illustration 4–1 QFD House of Quality for the VLA bimanual manipulation system.

Chapter 5 Concept generation and selection

5.1 Concept generation

5.1.1 Functional decomposition

The target assembly system was first decomposed into solution-neutral functions: establish the workspace state; create a stable robot–operator interface for demonstrations; record synchronized visual, proprioceptive, and action data; grasp and constrain the component during contact; map the current observation to coordinated motion; and detect or recover from an invalid execution state. The decomposition converts the QFD priorities into five design modules. The robot platform provides repeatable bimanual motion for ES1, ES2, and ES4; the teleoperation module enables usable demonstrations; the data-collection module supports ES3–ES4 and learning reproducibility; the gripper realizes component contact; and the policy maps observations and state to actions. Failure monitoring and recovery remain cross-cutting functions because they act across the collection and deployment loop.

5.1.2 Structured brainstorming

The team used structured brainstorming to generate alternatives for each module before choosing a configuration. Candidate ideas were retained when they addressed at least one QFD-linked specification and did not violate laboratory safety, hardware access, or the project schedule. The session considered the available FR3 ecosystem alongside Flexiv and ALOHA-style platforms; direct joint-master, XR, and hand-held teleoperation; progressively richer observation schemas; compliant and rigid end-effectors; and policy families ranging from Diffusion Policy to ACT and future VLA models. This procedure deliberately separates the choice of a data-collection interface from the choice of a learning algorithm, since neither can be justified independently of the sensing and robot-control path.

The generated alternatives were also informed by established research directions rather than by ad hoc feature lists. Robomimic and MimicGen motivate treating demonstration quality and dataset construction as system-level design variables[robomimic,mimicgen]; RVT, 3D Diffusion Policy, and R3M respectively motivate the retained alternatives for multi-view geometry, 3D visuomotor control, and reusable robot visual representations[rvt,dp3,r3m]. These references define the external concept space, while the selected configuration remains constrained by the laboratory hardware, the QFD priorities, and the project schedule.

5.1.3 Morphological chart and generated module concepts

Table 6–1 records the resulting concept space. A row represents one functional module, and a column entry represents one candidate realization. The bold entry in each row is the selected module concept; the table is not a claim that the unselected entries were all built or experimentally benchmarked.

Table 5–1 Morphological chart for the five functional modules

Concept	Candidate A	Candidate B	Candidate C
C1: Platform	**Dual Franka FR3**	Dual Flexiv platform	ALOHA-style bimanual platform
C2: Teleoperator	**GELLO joint-master**	PICO 4 Ultra XR	UMI hand-held
C3: Data collection	**Two wrist + one top camera + robot state/action**	One wrist + one top camera + robot state/action	Robot state/action only
C4: Gripper	Franka parallel gripper	3D-printed hard gripper	**3D-printed soft gripper**
C5: Policy	**ACT in LeRobot**	π-style VLA policy	Diffusion policy

The morphological chart produces five leading module concepts, denoted C1–C5. C1 is a dual-FR3 platform; C2 is GELLO-based teleoperation; C3 is a synchronized three-camera, robot-state, and action pipeline; C4 is a 3D-printed soft gripper; and C5 is an ACT policy integrated through LeRobot. Together they form one integrated system concept, rather than five competing end-to-end robot systems. Their alternatives are retained because each exposes a real engineering trade-off: platform precision against integration cost, operator intuitiveness against mapping complexity, observation coverage against synchronization burden, compliance against geometric repeatability, and policy capability against data and deployment maturity. Detailed descriptions of the unselected alternatives are provided in the appendix.

5.2 Concept selection

5.2.1 Selection method and QFD-derived criteria

Selection was conducted in two stages. First, an absolute screen removed a candidate if it could not be made safe in the laboratory, could not interface with the selected robot and data path, or could not be obtained and integrated within the project schedule. Second, the surviving alternatives were scored on a five-point ordinal scale, where 1 denotes a weak fit and 5 denotes a strong fit. Ratings were assigned from the QFD relationship to ES1–ES6, laboratory availability and interface knowledge, the existing collection workflow, and the cited primary or official sources; no rating is presented as an unreported benchmark result. Consistent with the functional decomposition, the "top five concepts" are the five selected module concepts C1–C5. Their combination forms the single system concept carried into the final design. For candidate a in module m, the decision score is

$$S_{m,a} = \sum_{k=1}^{4} w_k r_{m,a,k}, \tag{5–1}$$

where $r_{m,a,k}$ is the rating for criterion k and w_k is its QFD-derived weight. Scores are used only to compare alternatives within the same module; they are recorded design judgments, not measured task-performance results.

Table 5–2 QFD-derived criteria for module selection

No.	Criterion	Weight	QFD Link
SC1	Contribution to task performance and precision	0.30	ES1, ES2, ES3, ES4
SC2	Data, control, and software integration compatibility	0.25	ES1, ES2, ES3
SC3	Safety, robustness, and recovery support	0.20	ES5, ES6; CR3–CR4
SC4	Availability, manufacturability, and schedule feasibility	0.25	CR6–CR7 and project plan

5.2.2 Scoring of the five selected module concepts

Table 6–3 provides the detailed selection record for the five leading module concepts. The same criterion definitions and weights are applied in every module so that the scoring logic remains traceable to the QFD; however, the totals are interpreted only within a module because a robot platform, a gripper, and a policy make different contributions to the integrated system.

Table 5–3 Module-level concept selection matrix (1 = weak fit; 5 = strong fit)

Module	Candidate	SC1	SC2	SC3	SC4	Weighted score
C1 Platform	Dual Franka FR3	5	5	4	3	**4.30**
	Dual Flexiv	4	3	4	2	3.30
	ALOHA-style bimanual	3	4	3	5	3.85
C2 Teleoperator	GELLO joint-master	4	5	4	5	**4.50**
	PICO 4 Ultra XR	3	3	3	4	3.25
	UMI hand-held	3	3	3	4	3.25
C3 Data collection	Two wrist + top camera + state/action	5	5	4	3	**4.35**
	One wrist + top camera + state/action	4	4	4	5	4.20
	Robot state/action only	2	3	3	5	3.25
C4 Gripper	3D-printed soft gripper	4	4	4	5	**4.20**
	Franka parallel gripper	3	5	4	3	3.70
	3D-printed hard gripper	4	5	3	4	4.05
C5 Policy	ACT in LeRobot	4	5	4	5	**4.50**
	Diffusion Policy	4	3	4	4	3.75
	π-style VLA policy (future route)	5	2	3	1	2.75

5.2.3 Comparative evaluation and selected concept

C1: Dual Franka FR3 platform. The dual Franka FR3 platform was selected as the mechanical foundation of the final system because it provides repeatable 7-DoF control, a direct connection to the available laboratory control ecosystem, and a credible path toward coordinated bimanual manipulation. Its principal disadvantage is the substantial integration effort required for workspace coordination, collision avoidance, commissioning, and bilateral control. Nevertheless, these costs are justified by the project objective: lower-cost alternatives may reduce initial hardware burden, but they would introduce a different control ecosystem or provide a weaker fit to the required assembly precision.

C2: GELLO teleoperation. GELLO was selected as the planned demonstration-collection interface because it offers a direct and intuitive connection between operator motion and the robot-state/action records required by the learning pipeline[gello]. Despite the need for reliable leader–follower mapping and calibration, GELLO remains preferable to immersive or hand-held alternatives for the present system because it provides the most direct route from teleoperated demonstrations to the joint-space data representation used by the FR3 and LeRobot workflow.

C3: Three-camera data pipeline. The selected sensing configuration uses one Intel RealSense D435 overhead camera and two Intel RealSense D405 wrist-mounted cameras, together with synchronized robot state, gripper state, and action records. The D435 provides a stable global view of the shared workspace, while the two D405 cameras preserve local visual information near the end-effectors and contact regions. This complementary arrangement reduces visual blind spots and directly supports the component-detection and pose-estimation requirements in ES3–ES4, at the cost of increased synchronization and data-management effort.

C4: 3D-printed soft gripper. The selected end-effector is a 3D-printed soft gripper derived from the open-source gripper design in the referenced UMI material; the project adopts only the gripper-finger component rather than reproducing the complete UMI hardware system[umi]. Compliant contact and rapid geometry iteration are valuable for handling assembly components with small geometric variation, although deformation may reduce pose repeatability compared with a rigid locating gripper.

C5: ACT policy in LeRobot. ACT was selected as the current policy because it is integrated with the LeRobot workflow and supports task-specific training and checkpoint deployment from synchronized visual observations and robot-state inputs[act,lerobot]. Action chunking provides a practical temporal-control mechanism for the present imitation-learning pipeline. Deployment remains task-specific and operator-selected: the policy does not yet infer tasks from natural-language input or demonstrate unified multi-task control. This limitation is accepted because ACT provides a deployable and traceable baseline for the current project stage, whereas Diffusion Policy remains an earlier baseline and π-style VLA models remain a future extension.

The selected architecture is the combination of the highest-scoring surviving candidate in each module. C1 and C4 establish the physical capability for controlled contact; C2 and C3 make demonstrations and visual evidence available to the learner; and C5 provides the deployable policy path. This combined decision covers the full ES1–ES6 chain while retaining a feasible implementation route for subsequent final design, manufacturing, and validation.

Chapter 6 Concept generation and selection

6.1 Concept generation

6.1.1 Functional decomposition

The target assembly system was first decomposed into solution-neutral functions: establish the workspace state; create a stable robot–operator interface for demonstrations; record synchronized visual, proprioceptive, and action data; grasp and constrain the component during contact; map the current observation to coordinated motion; and detect or recover from an invalid execution state. The decomposition converts the QFD priorities into five design modules. The robot platform provides repeatable bimanual motion for ES1, ES2, and ES4; the teleoperation module enables usable demonstrations; the data-collection module supports ES3–ES4 and learning reproducibility; the gripper realizes component contact; and the policy maps observations and state to actions. Failure monitoring and recovery remain cross-cutting functions because they act across the collection and deployment loop.

6.1.2 Structured brainstorming

The team used structured brainstorming to generate alternatives for each module before choosing a configuration. Candidate ideas were retained when they addressed at least one QFD-linked specification and did not violate laboratory safety, hardware access, or the project schedule. The session considered the available FR3 ecosystem alongside Flexiv and ALOHA-style platforms; direct joint-master, XR, and hand-held teleoperation; progressively richer observation schemas; compliant and rigid end-effectors; and policy families ranging from Diffusion Policy to ACT and future VLA models. This procedure deliberately separates the choice of a data-collection interface from the choice of a learning algorithm, since neither can be justified independently of the sensing and robot-control path.

The generated alternatives were also informed by established research directions rather than by ad hoc feature lists. Robomimic and MimicGen motivate treating demonstration quality and dataset construction as system-level design variables[robomimic,mimicgen]; RVT, 3D Diffusion Policy, and R3M respectively motivate the retained alternatives for multi-view geometry, 3D visuomotor control, and reusable robot visual representations[rvt,dp3,r3m]. These references define the external concept space, while the selected configuration remains constrained by the laboratory hardware, the QFD priorities, and the project schedule.

6.1.3 Morphological chart and generated module concepts

Table 6–1 records the resulting concept space. A row represents one functional module, and a column entry represents one candidate realization. The bold entry in each row is the selected module concept; the table is not a claim that the unselected entries were all built or experimentally benchmarked.

Table 6–1 Morphological chart for the five functional modules

Concept	Candidate A	Candidate B	Candidate C
C1: Platform	**Dual Franka FR3**	Dual Flexiv platform	ALOHA-style bimanual platform
C2: Teleoperator	**GELLO joint-master**	PICO 4 Ultra XR	UMI hand-held
C3: Data collection	**Two wrist + one top camera + robot state/action**	One wrist + one top camera + robot state/action	Robot state/action only
C4: Gripper	Franka parallel gripper	3D-printed hard gripper	**3D-printed soft gripper**
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Chapter 7 Final design and implementation

7.1 Selected concept description

7.1.1 Design evolution from the DR3 baseline

The final implementation retains the modular architecture established during Design Review 3 (DR3), but updates both the learned policy and the demonstrated task scope. DR3 used task-specific Action Chunking with Transformers (ACT) checkpoints as the deployable baseline and treated language-conditioned bimanual control as future work. During subsequent development, the policy was replaced by the LeRobot implementation of $\pi_{0.5}$, the state and action interfaces were expanded to two robot arms, and three long-horizon tasks were combined in one multi-task training run. ACT is therefore retained as a development baseline rather than the final deployed controller.

The resulting system is a language-conditioned bimanual manipulation platform. It combines a dual-Franka Research 3 (FR3) workcell, GELLO-based demonstration collection, three synchronized RGB cameras, bilateral robot and gripper state acquisition, a LeRobot Dataset v3.0 dataset, one jointly trained $\pi_{0.5}$ checkpoint, and robot-side execution and safety interfaces. $\pi_{0.5}$ is a vision-language-action policy designed to combine visual observations, natural-language task descriptions, robot state, and action generation[pi05]. The implementation used in this project follows the Pi05 policy integration in LeRobot, which is adapted from the open-source OpenPI implementation[lerobot,lerobotpi05,openpi].

7.1.2 Selected concept overview

The selected concept connects demonstration collection, dataset construction, policy training, real-robot deployment, and task-level validation in one closed pipeline. The physical workcell consists of two FR3 arms operating in a shared, bounded workspace. Two Intel RealSense D405 cameras provide local wrist views, and one Intel RealSense D435i provides a fixed global view from above. The workcell uses the custom compliant gripper developed during DR3. Its two identical screw-mounted fingers are 3D-printed in TPU 64D and have an overall bounding-box size of

$$123.36 \times 25.80 \times 40.00 \text{ mm}.$$

The compliant finger geometry accommodates variation in object shape and reduces concentrated contact forces, while the reinforced mounting region provides a repeatable mechanical interface.

Human demonstrations are collected through the GELLO leader devices[gello]. During collection, the system records the three camera streams, the current state of both arms and grippers, the commanded bilateral action, and the natural-language task instruction. Accepted episodes are stored in LeRobot Dataset v3.0 format and are checked for complete observation, state, action, video, and task fields before training.

At deployment time, the three tasks use one common $\pi_{0.5}$ checkpoint. The current camera observations, bilateral robot state, and selected language instruction are passed to the policy. The policy predicts an action chunk, of which the first configured execution segment is sent through the robot-side interface. The interface validates the action representation before sending absolute joint targets and absolute gripper-width targets to the two FR3 controllers. Joint limits, workspace limits, controller protections, and emergency-stop procedures remain outside the learned policy and are active throughout execution[frankafci].

7.1.3 Operational workflow

For each trial, the two FR3 arms, grippers, cameras, task fixture, and manipulated objects are returned to a documented initial condition. The operator then selects one of the three task instructions and verifies that the instruction matches the prepared scene. The policy receives the instruction together with the synchronized visual and proprioceptive observations and generates the bilateral action sequence. After each executed segment, new observations and robot states are acquired and the policy is queried again.

The three tasks exercise complementary forms of long-horizon bimanual manipulation:

1. repositioning a can, grasping a handled ring, and placing the ring over the can;
2. repositioning a socket box, grasping a plug, and performing constrained plug insertion; and
3. repositioning a motherboard, extracting a memory module, and placing it in a tray.

The common checkpoint and language interface allow the requested behavior to be changed without replacing the deployed policy file.

7.2 Engineering design analysis

7.2.1 Bimanual state and action representation

The recorded state and the policy action both use a 16-dimensional `float32` representation. The state is

$$\mathbf{s}_t = \left[q^L_{1,t}, \dots, q^L_{7,t}, g^L_t, q^R_{1,t}, \dots, q^R_{7,t}, g^R_t\right]^\mathsf{T} \in \mathbb{R}^{16}, \tag{7–1}$$

where $q^L_{1,t}, \dots, q^L_{7,t}$ and $q^R_{1,t}, \dots, q^R_{7,t}$ are the measured joint positions of the left and right FR3 arms, and g^L_t and g^R_t are the measured gripper opening widths in metres.

The action has the same ordering:

$$\mathbf{a}_t = \left[q^{L,\text{target}}_{1,t+1}, \dots, q^{L,\text{target}}_{7,t+1}, g^{L,\text{target}}_t, q^{R,\text{target}}_{1,t+1}, \dots, q^{R,\text{target}}_{7,t+1}, g^{R,\text{target}}_t\right]^\mathsf{T}. \tag{7–2}$$

The arm actions are absolute joint-position targets for the next control step, rather than delta actions. The gripper actions are absolute opening-width targets, `open_width_m`, expressed in metres. Indices 0–6 correspond to the left arm, index 7 to the left gripper, indices 8–14 to the right arm, and index 15 to the right gripper. The configuration therefore uses `use_relative_actions=false` for both training and deployment.

The $\pi_{0.5}$ implementation pads the state and action features to its maximum internal dimension of 32. Only the first 16 entries contain valid project data; the remaining entries are padding and are not sent as additional robot commands. This distinction is maintained in the dataset feature specification and robot action adapter.

7.2.2 Temporal interface and action chunking

Data collection, policy observations, and robot control operate at 5 Hz, so one step represents 0.2 s. The model predicts a chunk of 50 future actions, corresponding to a nominal prediction horizon of

$$T_\text{chunk} = \frac{50}{5} = 10\ \text{s}.$$

Deployment uses `n_action_steps=10`; therefore, ten actions, or approximately 2 s of robot motion, are consumed before a new policy query. Receding-horizon execution allows the next prediction to incorporate updated visual observations and bilateral robot state instead of executing the full ten-second prediction open loop.

7.2.3 Perception and camera configuration

The final visual configuration contains three RGB cameras. One fixed Intel RealSense D435i provides an overhead view of the shared workcell, fixtures, and large object displacements. Two Intel RealSense D405 cameras provide left- and right-wrist views of grasping and contact regions. This division preserves global scene context while reducing the loss of local detail caused by occlusion during insertion, extraction, and placement.

All three streams are recorded at 424×240 pixels (width $\times$ height). In the dataset, each image is represented in channel-first format as

$$[C, H, W] = [3, 240, 424].$$

Before an image is passed to $\pi_{0.5}$, it is converted to $3 \times 224 \times 224$. Preprocessing preserves the original aspect ratio: the 424×240 image is scaled to approximately 224×126, and approximately 49 pixels of black padding are added above and below the resized image. The image is not stretched into a square. Pixel values are first represented in $[0, 1]$ and are then normalized to $[-1, 1]$ before entering the model. The deployed policy therefore receives three tensors of shape $3 \times 224 \times 224$ at each observation step.

Camera identity and preprocessing are part of the policy feature contract. Before recording or deployment, the system checks that the overhead, left-wrist, and right-wrist streams are present and mapped to the same logical feature names used during training. A missing, stale, or incorrectly mapped stream causes the run to stop rather than allowing the policy to act on an unintended camera ordering.

7.2.4 Packet synchronization and action-label alignment

The data stream is sampled at 5 Hz, corresponding to an interval of approximately 0.2 s. The recorder pairs adjacent assembled packets rather than interpolating raw device timestamps. For the saved sample at index t,

$$\mathbf{o}_t^{\text{visual}} = \text{three images from packet } t, \tag{7–3}$$

$$\mathbf{s}_t = \text{bilateral state from packet } t, \tag{7–4}$$

$$\mathbf{a}_t^{\text{arm}} = \mathbf{q}_{t+1}^{\text{target}} \text{ from packet } t+1, \tag{7–5}$$

$$\mathbf{a}_t^{\text{gripper}} = \mathbf{g}_t^{\text{target}} \text{ from packet } t. \tag{7–6}$$

Thus, the three images and the measured robot state come from the same `current_packet`. The supervised arm label is shifted forward by one packet so that the observation at t is paired with the next absolute joint target. The gripper label follows the existing same-packet convention and stores the absolute commanded opening width from packet t.

This procedure establishes software-level synchronization within one packet. The recorder does not perform device-timestamp interpolation, ROS ApproximateTime alignment, or hardware-triggered exposure synchronization. It relies on the upstream ROS 2 bridge to assemble the packet. Consequently, the dataset supports the claim that camera images and robot state share one software packet, but it does not by itself prove that the three camera exposures occurred at exactly the same physical instant.

7.2.5 Multi-task dataset construction

Each of the three tasks contains 80 episodes, for a total of 240 episodes and 119,650 frames. All tasks use random seed 1000 and the same held-out episode indices:

$$\{17, 35, 58, 65, 66, 72, 74, 75\}.$$

The remaining 72 episodes from each task are used for training. The split is performed by complete episode, preventing frames from one physical trajectory from appearing in both the training and validation sets.

During joint training, tasks are sampled with equal probability. The sampler first selects the task and then selects training data from that task, so the larger number of frames in `dual_ring` does not give it a larger task-level sampling probability. This preserves balanced supervision across the three language instructions despite their different trajectory lengths.

Table 7–1 Episode-level dataset split for the three bimanual tasks

Task dataset	Episodes	Frames	Training split	Validation split
`dual_ring`	80	46,326	72 episodes / 41,410 frames	8 episodes / 4,916 frames
`dual_socket`	80	38,511	72 episodes / 34,655 frames	8 episodes / 3,856 frames
`dual_ram`	80	34,813	72 episodes / 31,675 frames	8 episodes / 3,138 frames
Total	**240**	**119,650**	**216 episodes / 107,740 frames**	**24 episodes / 11,910 frames**

7.2.6 LeRobot $\pi_{0.5}$ policy

The final controller uses one jointly trained $\pi_{0.5}$ checkpoint. The base checkpoint is the official `pi05_base` model, whose upstream parameters are identified by gs://openpi-assets/checkpoints/pi05_base/params. Training reads the converted PyTorch weights from real-exp/.hf-cache/pi05_base_pytorch/model.safetensors. The base model combines a PaliGemma `gemma_2b` vision-language component with a `gemma_300m` action expert. Model weights use `bfloat16`.

The implementation uses LeRobot 0.5.2 at commit `dc06dba9d6220ab99c94e83ec7f0d45a35b` together with project-specific local modifications. Training is performed on physical GPU 1 of an NVIDIA RTX A6000 with 48 GB of memory. The three tasks are trained jointly for 100,000 optimizer steps with batch size 4 and AdamW. The peak learning rate is 2.5×10^{-5}. A 1,000-step warm-up is followed by cosine decay to 2.5×10^{-6}. Validation is run every 5,000 steps.

Checkpoint selection uses a task-balanced validation quantity rather than a frame-weighted loss. For each task, the mean normalized flow-matching loss is first calculated over at most 64 validation batches. With batch size 4, this evaluates at most 256 validation samples per task at each validation point. The checkpoint-selection metric is

$$\mathcal{L}_{\text{macro}} = \frac{\mathcal{L}_{\text{ring}} + \mathcal{L}_{\text{socket}} + \mathcal{L}_{\text{ram}}}{3}, \tag{7–7}$$

where each term is the corresponding task's mean normalized flow-matching loss. The best checkpoint is the validation point with the lowest $\mathcal{L}_{\text{macro}}$. The unweighted arithmetic mean prevents the longer dual_ring trajectories from dominating checkpoint selection.

Table 7–2 Final $\pi_{0.5}$ training and deployment configuration

Parameter	Implemented value
Base model	Official pi05_base
Model structure	PaliGemma gemma_2b + action expert gemma_300m
Weight precision	bfloat16
LeRobot version	0.5.2
LeRobot commit	dc06dba9d6220ab99c94e83ec7f0d45a35b44bed, with local modifications
Dataset format	LeRobot Dataset v3.0
Training hardware	NVIDIA RTX A6000 48 GB, physical GPU 1
Batch size	4
Optimizer	AdamW
Training length	100,000 steps across three jointly sampled tasks
Learning-rate schedule	1,000-step warm-up to 2.5×10^{-5}, cosine decay to 2.5×10^{-6}
Validation interval	Every 5,000 steps
Checkpoint-selection metric	Lowest macro-average normalized flow-matching loss across the three tasks; at most 64 batches per task
Action chunk	50 predicted steps
Executed segment	n_action_steps=10, approximately 2 s at 5 Hz
Action representation	Absolute joint and gripper-width targets; use_relative_actions=false

7.2.7 Control, computation, and safety

The learned model provides high-level bilateral targets, while the robot-side interface and FR3 controllers remain responsible for admissibility checks and low-level execution. Before motion is enabled, a dry run verifies the three camera streams, the 16-dimensional state and action mapping, finite policy outputs, checkpoint and dataset statistics, language instruction, communication status, joint limits, workspace bounds, gripper limits, and emergency-stop access.

During execution, a missing observation, stale state, non-finite output, communication timeout, excessive joint target, workspace violation, protective stop, or manual emergency stop terminates the run. The power socket remains electrically unpowered. Reset is performed only after policy execution has stopped and the shared workspace is clear. These controls prevent the VLA policy from bypassing deterministic safety constraints.

7.3 Detailed system design

7.3.1 Dual-arm robotic platform

The workcell uses one FR3 arm on each side of the shared workspace. Both arms provide seven joint-position channels and one gripper-width channel to the learned policy. The bimanual arrangement enables one arm to prepare or stabilize the scene while the other performs grasping, insertion, extraction, or placement. The recorded demonstrations determine the coordinated timing between the arms.

7.3.2 Task-compatible gripper

The final workcell uses the custom gripper developed during DR3. The compliant TPU fingers are particularly useful for the plug body and handled ring because they tolerate small shape and pose variations. For memory-module removal, the same gripper concept must maintain sufficient geometric access to the module while avoiding contact with neighboring motherboard components. The gripper is inspected before each run for finger damage, loose screws, excessive deformation, and insufficient opening range.

7.3.3 Hardware–software interfaces

7.3.4 Task definitions and bilateral role allocation

Table 7–4 records the exact task-level language instructions, the demonstrated allocation of work between the arms, and the terminal completion conditions.

7.4 Manufacturing and implementation

7.4.1 Implementation workflow

Implementation proceeded through five linked stages. First, the dual-FR3 workcell, custom grippers, fixtures, cameras, and safety controls were commissioned. Second, the dual GELLO interfaces and robot bridges were used to record 80 demonstrations for each language instruction. Third, the episodes were checked, divided by episode into the fixed training and

Table 7–3 Principal interfaces in the final dual-arm VLA system

Interface	Connected modules	Transferred information
Teleoperation interface	GELLO leaders to two FR3 arms	Bilateral joint and gripper targets for demonstration collection
Visual interface	D435i overhead and two D405 wrist cameras to LeRobot	Three synchronized RGB observations
Robot-state interface	Two FR3 controllers and grippers to LeRobot	Sixteen valid state values in the fixed left-to-right ordering
Dataset interface	Recorder to LeRobot Dataset v3.0	Images, state, absolute action, task instruction, episode index, and dataset statistics
Policy interface	LeRobot dataset and language input to $\pi_{0.5}$	Three-task balanced training input and 50-step action prediction
Execution interface	$\pi_{0.5}$ to robot-side adapter	First ten absolute bilateral targets from each predicted action chunk
Safety interface	Robot-side checks and FR3 protections	Joint, gripper, workspace, freshness, finite-value, timeout, and stop conditions

Table 7–4 Language instructions, bilateral roles, and completion conditions

Task	Language instruction	Demonstrated arm roles	Terminal completion condition
T1	First, push the can to the middle, then pick up the black ring with the handle, put it over the can, and set it down.	Left arm pushes the can; right arm grasps the handled ring and places it.	The can is repositioned, and the released ring remains stably over the can.
T2	Push the box with the power socket to the middle, then pick up the plug and insert it into the socket.	Right arm pushes the socket box; left arm grasps and inserts the plug.	The socket box is repositioned, and the plug remains inserted without manual assistance.
T3	Pull the computer motherboard closer, then remove the memory module and place it in the yellow tray.	Left arm grips and pulls the motherboard; right arm removes and places the memory module.	The module is fully removed and remains inside the yellow tray.

validation sets, and converted to LeRobot Dataset v3.0. Fourth, the official `pi05_base` weights were converted and fine-tuned through one balanced 100,000-step training run. Finally, the resulting checkpoint was connected to the three-camera, 16-dimensional state, and bilateral absolute-action interfaces and tested through dry runs before physical execution was enabled.

7.4.2 Deployment procedure

Each deployment uses the following sequence:

1. secure the fixtures and task objects and clear the shared workspace;
2. verify the two FR3 arms, custom grippers, emergency-stop devices, and three camera streams;
3. load the unified checkpoint, dataset statistics, feature mapping, and action adapter;
4. confirm the 16-dimensional bilateral state and action ordering;
5. provide the language instruction that matches the prepared task;
6. run policy inference with physical motion disabled and inspect the first action segment;
7. enable the robot-side executor and begin the counted trial; and
8. stop on success, timeout, invalid observation or action, protective stop, emergency stop, or manual intervention.

7.4.3 Current implementation status

The current implementation uses one LeRobot $\pi_{0.5}$ checkpoint with two FR3 arms and has completed the end-to-end integration path for the ring-over-can, plug-in-socket, and memory-module-removal tasks. This establishes that the language input, three visual streams, bilateral state and action interfaces, policy inference, receding-horizon execution, and robot controllers can operate together on the physical workcell. It is an implementation claim; quantitative performance claims require the repeated-trial counts, execution times, failure inventory, and uncertainty estimates reported in Chapter 9.

Chapter 8 Validation results

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8.1 Validation overview

The validation procedure follows the DR3 engineering specifications in Chapter 4. The purpose of validation is to check whether the integrated system satisfies the intended proof-of-concept behavior under bounded laboratory conditions. The evaluated workflow includes scene setup, synchronized sensing, state acquisition, policy inference, robot and gripper command execution, and completion assessment.

The bimanual manipulation task has been completed on the integrated dual-arm setup. This result demonstrates that the selected hardware platform, teleoperation-derived data path, ACT-based policy module, and robot-side execution loop can be combined into a functioning bimanual manipulation system. The result should be interpreted within the defined project scope: the workspace, initial conditions, fixtures, and execution assumptions are controlled, and the system is not claimed to provide unrestricted open-world autonomy.

8.2 Key data

8.3 Validation summary

The completed validation confirms the main system-level objective of the project: a dual-arm FR3 manipulation platform can execute the scoped bimanual manipulation workflow using the selected data and policy architecture. The most important validation outcome is the integration of the complete chain from demonstration collection and dataset preparation to learned policy execution on the real robot. Detailed task descriptions, trial counts, and quantitative measurements are reserved for the blank key-data section.

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Chapter 10 Discussion and conclusions

10.1 Discussion

This project developed a bimanual industrial manipulation system for representative 3C assembly operations. The final system integrates a dual-Franka Research 3 platform, GELLO-based demonstration collection, synchronized visual and proprioceptive data recording, task-compatible end effectors and fixtures, and an ACT-based policy deployment workflow. The design process remains traceable from customer requirements to engineering specifications, concept generation, concept selection, implementation, and validation.

A key outcome is that the bimanual manipulation task has been completed on the integrated real-robot setup. This confirms that the selected mechanical, sensing, data, and policy modules can operate together as a system rather than only as isolated components. The completed bimanual result also supports the original motivation of moving beyond single-arm demonstrations toward coordinated manipulation in a shared workcell.

The main limitation is that the present implementation remains a bounded laboratory proof of concept. The task setup, workspace, initial conditions, and fixtures are controlled. The ACT controller provides a deployable imitation-learning baseline, while language-conditioned checkpoint selection, unified multi-task VLA inference, autonomous planning, and generalized recovery remain outside the demonstrated scope.

10.2 Conclusions

The thesis presents an end-to-end engineering process for VLA-oriented bimanual industrial manipulation. Starting from customer requirements, the project defined measurable engineering specifications, generated modular solution concepts, selected a dual-FR3 and GELLO-based architecture, and implemented a data-driven manipulation pipeline using synchronized sensing and ACT policy deployment.

The completed system demonstrates that a dual-arm robot workcell can be integrated with teleoperation, dataset generation, learning-based control, and real-robot execution for the scoped bimanual manipulation task. The contribution is therefore both a working prototype and a traceable design methodology: the final design choices can be related back to the QFD requirements, while the validation structure provides a basis for reporting task success, perception quality, pose accuracy, failure handling, and recovery once the detailed data section is completed.

Overall, the project establishes a practical foundation for flexible 3C assembly research. It shows that bimanual manipulation can be achieved within a controlled laboratory workcell using the selected hardware and learning pipeline, while leaving task-specific details and quantitative key data to be documented separately.

Chapter 11 Discussion and conclusions

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